



NeuraSign

Physiological intelligence for work

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PROBLEM

Work is dynamic. So is human capacity.

WHAT COMPANIES KNOW

Skills



Role



Calendar



Deadlines



THE MISSING CONTEXT

Fatigue



Focus



Stress



Cognitive load



Readiness



The risk: **lost time** and **costly errors**.

Amazon

\$56M

Jury award

Loss of alertness.
Head-on collision.

Amazon delivery partner driver

Crash 16 Nov 2024

Verdict Jul 2026

INSURANCE JOURNAL

Jury Awards 78-Year-Old Victim \$56 Million for Crash Caused by Amazon Delivery Driver

By Andrew G. Simpson | July 27, 2026



Article photo: Amazon

Dry Tortugas

\$5M

Vessel and platform damage

Extra workload.

Asleep at the helm.

Reported sleep: 8–9 hours a day

Collision 16 Aug 2025

NTSB report 26 Aug 2026

NATIONAL TRANSPORTATION SAFETY BOARD



Left to right: The *Dry Tortugas* underway in January 2024, and platform VK-817A at an unknown date after the casualty. (Sources: Scott Rambo, marinetraffic.com, and Third Coast)

Contact of Offshore Supply Vessel Dry Tortugas with Offshore Platform VK-817A

Same skills. Different capacity.

Same person, different moments



 Lower cognitive load

 Higher cognitive load



Physiological signals

+ personal baseline

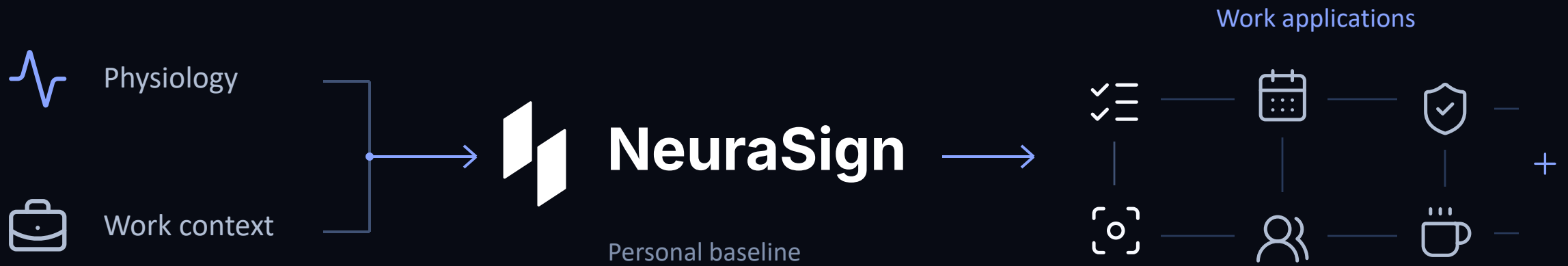


State estimates



Work decisions

Human-state intelligence for adaptive work.



Goal: adapt tasks, breaks and shifts to each person's capacity.

TARGET PRIORITY

01



**High-risk
operations**

02



**High-pressure
environments**

03



**High-performance
teams**

Stress and readiness

Classical supervised learning. No deep learning.

STRESS

94%

correct predictions

81 correct / 86 one-minute samples

Rest vs lab stress (WESAD)

ExtraTrees

300 decision trees

READINESS

3.98

points average error / 100

Average gap from Oura's daily score

4 unseen people / 684 test days

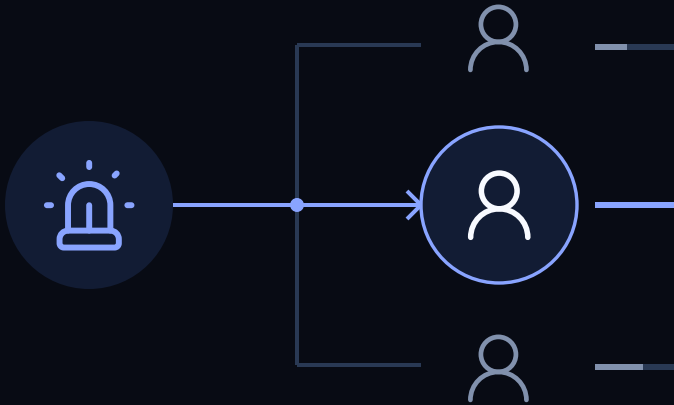
SVR + two CatBoost models

IFH Affect dataset

01

Dynamic task assignment

SOC + hospital emergencies

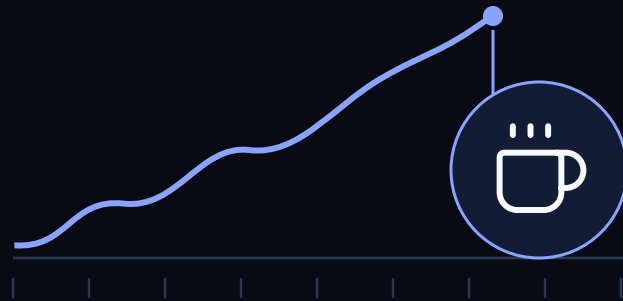


Suggest the next responder.

02

Overload prevention

Long hospital shifts



Prompt a break or rotation.

03

Safety-critical operations

Surgery



Support a pre-op team check-in.

Vision: combine physiology and work context to guide team decisions.



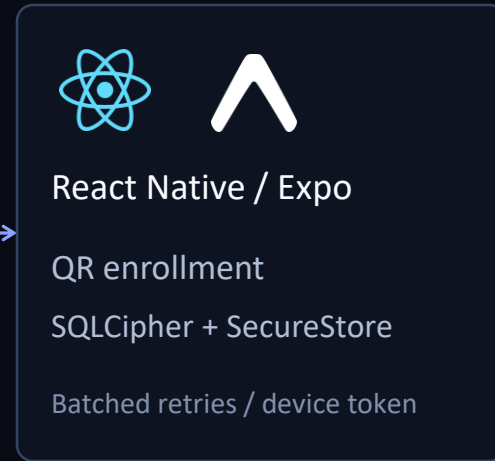
Planned advantage: workplace datasets tied to real outcomes and workflow integrations.

Qualitative product positioning. NeuraSign shows product vision.

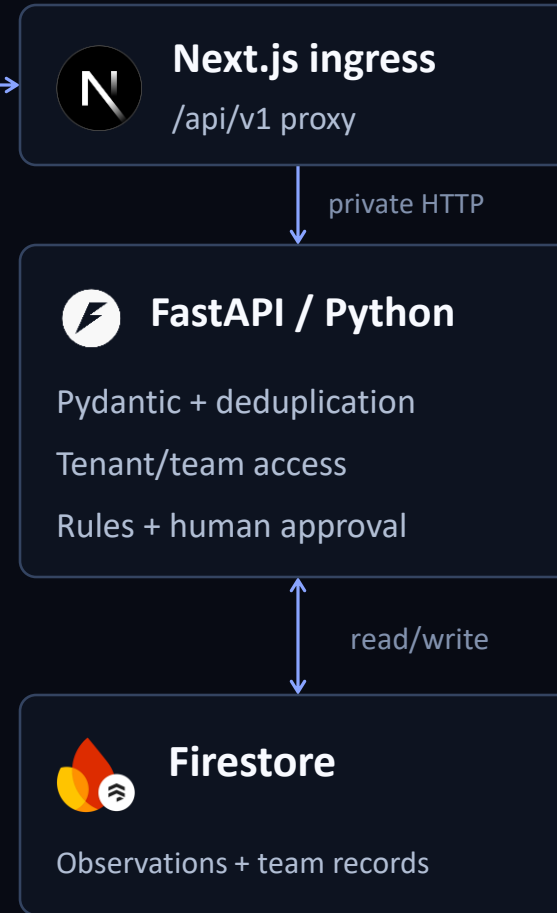
Wearables



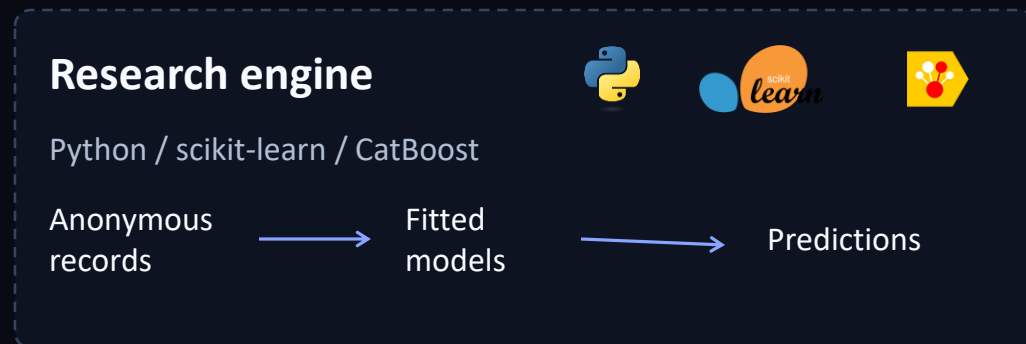
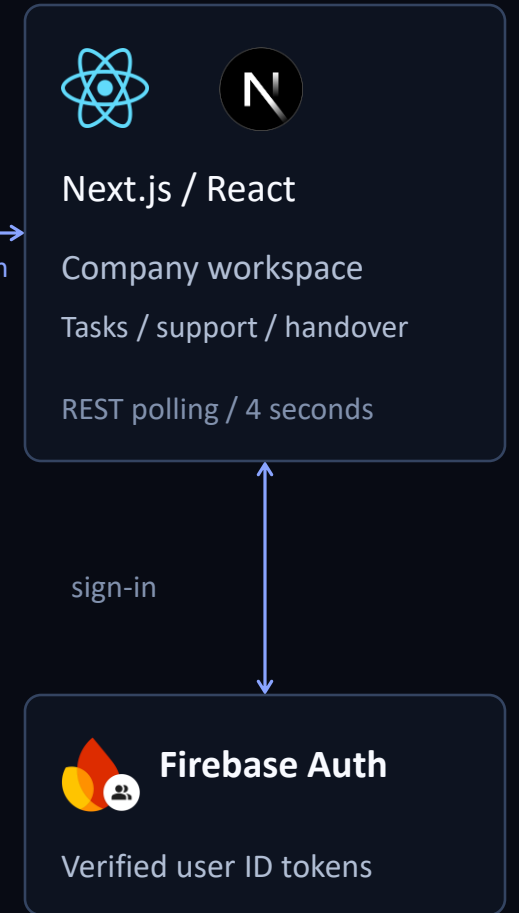
Phone app



Server



Dashboard



BLE

HTTPS

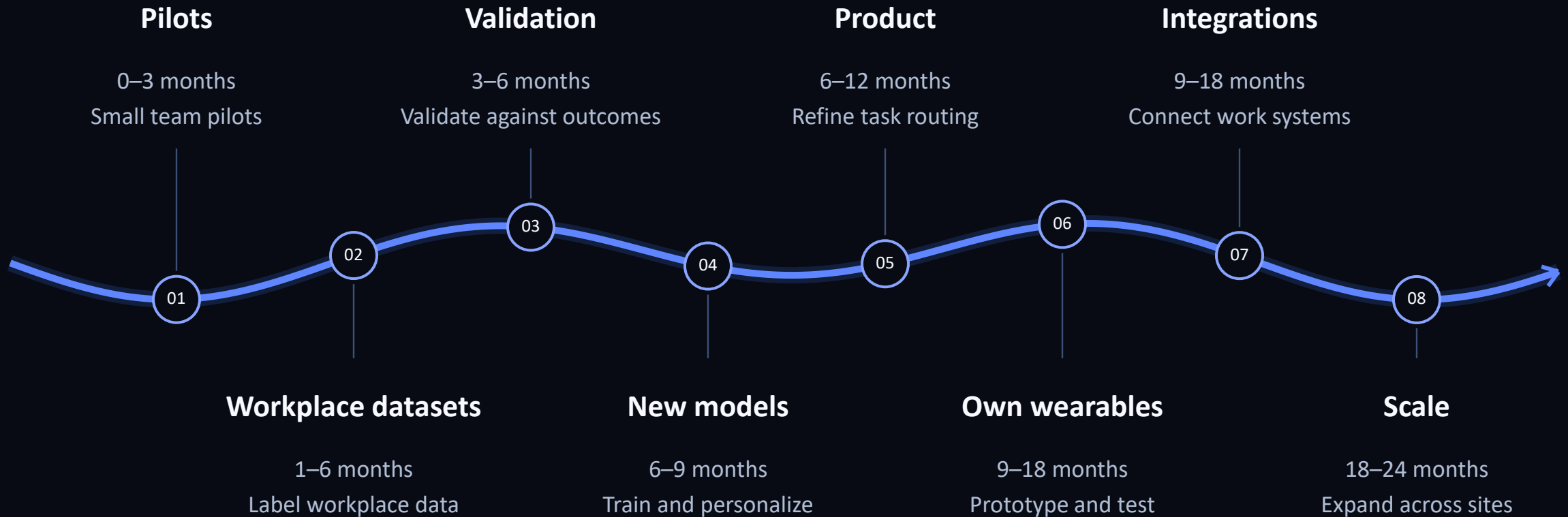
private HTTP

ID token

read/write

sign-in

Work that adapts to human capacity.



Proposed months from kickoff. Phases overlap.

Scientific evidence

Evidence for state estimates

 **Stress** HRV + EDA
Schmidt et al. (2018)

 **Mental fatigue** HR / HRV
Matuz et al. (2021)

 **Workload** HRV + EDA
Hirachan et al. (2022)

 **Readiness*** HRV
Kiviniemi et al. (2007)

* Indirect support from
physical-training research.

Market adoption



~300

EGA workers
Physiological wearables

Industrial heat safety
Summer 2024

Emirates Global Aluminium

Our proof

Lab stress / WESAD

94.2%

Selected-model accuracy
80% train / 20% test (people)
96.5% logistic reference

Daily Oura score

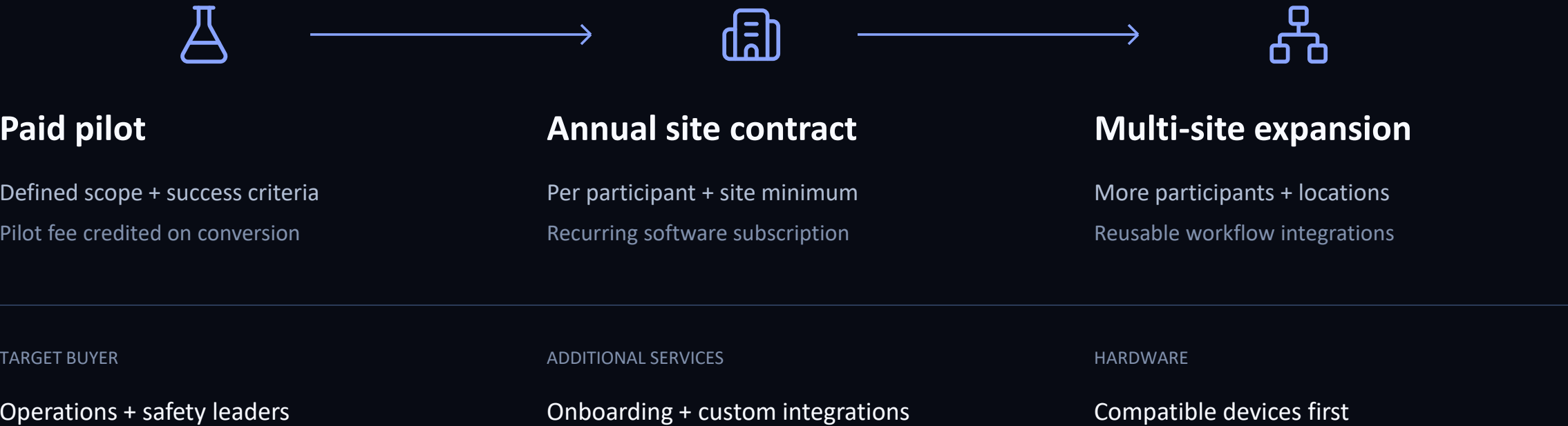
~4 points / 100

Reported prediction error

Offline research
Dashboard integration pending

Annual software. Per active participant.

State insights, work recommendations and workflow integrations



Proposed model. Pricing and contract terms to validate in pilots.

Control over data and decisions.

IMPLEMENTED IN THE PROTOTYPE

Pause & delete

Pause sharing or delete readings

Scoped access

Company + team permissions

Protected transfer

Encrypted queue + HTTPS

Managers see work context. Physiological data needs separate permission.

PROPOSED PILOT COMMITMENTS

Voluntary participation

No penalty for declining
or withdrawing

Purpose limits

Minimum necessary data
No ranking or disciplinary use

Accountable decisions

Human review and challenge
No fitness-for-work judgments

Before pilots: validate consent, retention and participant access to deletion.

Model evaluation

Offline tests with people separated across splits

Stress / WESAD

94.2%

accuracy / rest vs lab stress

96.5% logistic reference

3 held-out people / 60-second windows

Readiness / daily Oura score

~4

points error / 100

Approximation of a daily vendor score

Instant readiness remains unvalidated

Fatigue

63.7%

balanced accuracy

66.7% previous model / same evaluation

Reliability remains insufficient

Workload / NASA-TLX

~13

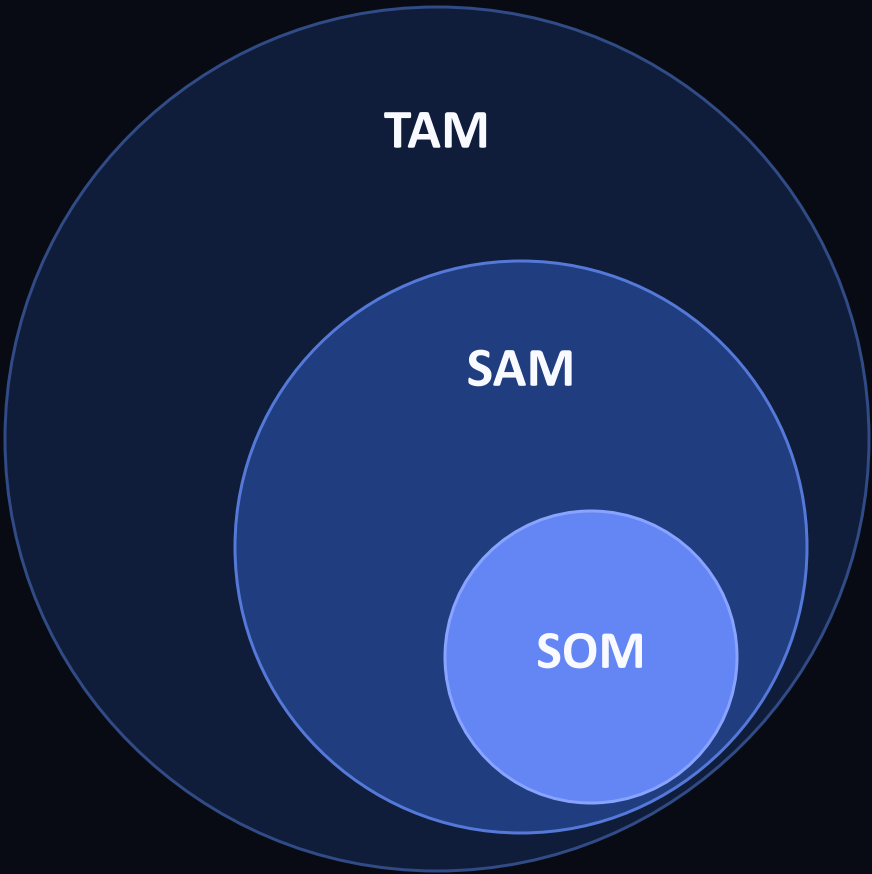
points error / 100

Similar error using only rest vs activity

Added value from physiology unproven

U.S. TAM / SAM / SOM

Illustrative pricing: \$20 per participant / month



BLS QCEW 2025 / Circles not to scale

SAM eligibility and SOM rollout are planning assumptions, not validated demand or a sales forecast.

TAM / SECTOR CEILING

\$4.9B / year

20.37M U.S. jobs × \$240/year
Manufacturing, transport, mining + utilities

SAM / ELIGIBILITY SCENARIO

\$244M / year

1.02M eligible seats × \$240/year
5% of sector jobs assumed eligible

SOM / INITIAL ROLLOUT

\$240k / year

10 sites × 100 participants × \$240/year
Illustrative 24-month deployment scenario

Where the products overlap.

Product	Scope	Primary inputs	Decision support
 FATIGUE SCIENCE	Workforce fatigue	Sleep + work schedules	Forecasts, breaks + reassignment
 SMART CAP Home Safe • Every Day	Operator alertness	EEG + operational data	Alerts, escalation + task rotation
 SOTER	Workplace safety	Motion + task video	Coaching + corrective actions
 CATAPULT	Sports teams	Location, motion + HR	Training load + return to play
 OURA	Organizations	Ring physiology + sleep	Group readiness + resourcing
 WHOOP [®]	Consumer	Physiology + training	Daily effort + recovery coaching
 NeuraSign	Product vision	Physiology + skills + task context	Work allocation + adaptive workflows

Vendor capabilities shown as documented. NeuraSign row describes the proposed product.

Sources: official product documentation, Sep 2026. Comparative effectiveness has not been established.

Supervised learning



WESAD / held-out test

12 development	3 held out
	
434 windows	57 features
60s per window	

32 configurations / 4 person-grouped folds

Selection uses development participants only.

Latest models / nested validation

Readiness / Fatigue / Workload

4x

outer folds



Inner folds select. Outer folds evaluate.

People remain separate in every split.